

3. Half the nerve cells in the average baby's brain die over the first few years of life.
4. Most children walk when they are ready, and no amount of encouragement will enable a 6-month-old to walk alone.
5. A person's hormones have little effect on human growth and development until puberty.
6. Emotional trauma can seriously impair the physical growth of young children, even those who are adequately nourished, free from illness, and not physically abused.

Jot down your responses and we will see how you did on this “pretest” as we discuss these issues throughout the chapter. (If you would like immediate feedback, the correct answers appear at the bottom of the page.)

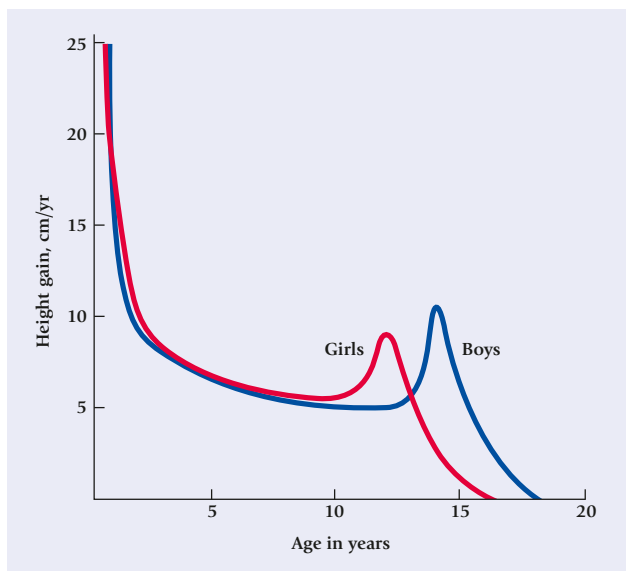
An Overview of Maturation and Growth

Adults are often amazed at how quickly children grow. Even tiny babies don't remain tiny for long: in the first few months of life, they gain nearly an ounce each day and an inch each month. Yet the dramatic increases in height and weight that we see are accompanied by a number of important *internal* developments in the muscles, bones, and central nervous system that will largely determine the physical feats that children are capable of performing at different ages. In this section of the chapter, we will briefly chart the course of physical development from birth through adolescence and see that there is a clear relationship between the external aspects of growth that are so noticeable and the internal changes that are much harder to detect.

Changes in Height and Weight

Babies grow very rapidly during the first 2 years, often doubling their birth weight by 4 to 6 months of age and tripling it (to about 21 to 22 pounds) by the end of the 1st year. Growth is very uneven in infancy. One study found that babies may remain the same length for days or weeks at a time before showing spurts of more than a centimeter in a single day (Lampl, Veldhuis, & Johnson, 1992). By age 2, toddlers are already half their eventual adult height and have quadrupled their birth weight to 27 to 30 pounds. If children continued to grow at this rapid pace until age 18, they would stand at about 12 feet 3 inches and weigh several tons!

From age 2 until puberty, children gain about 2 to 3 inches in height and 6 to 7 pounds in weight each year. During middle childhood (ages 6 to 11), children may seem to grow very little; over an entire year, gains of 2 inches and 6 pounds are hard to detect on a child who stands 4 to 4½ feet tall and weighs 60 to 80 pounds (Eichorn, 1979). But as shown in ■ Figure 5.1, physical growth and development are once again obvious at puberty, when adolescents enter a 2- to 3-year growth spurt, during which they may post an annual gain of 10 to 15 pounds and 2 to 4 inches in height. After this large growth spurt, there are typically small increases in height until full adult stature is attained in the mid- to late teens (Tanner, 1990).



■ **Figure 5.1** Gain in height per year by males and females from birth through adolescence. At about age 10½, girls begin their growth spurt. Boys follow some 2½ years later and grow faster than girls once their growth begins. *Based on a figure in Archives of the Diseases in Childhood, 41, by J. M. Tanner, R. H. Whithouse, and A. Takaishi, 1966, pp. 454–471.*

Answers to pretest: 1. F; 2. T; 3. T; 4. T; 5. F; 6. T